Gaming Center Network

Cisco Packet Tracer - Step-by-Step Build Guide (Project 3)

Goal: build a structured network for a gaming center with 16 gaming PCs: separate VLANs for gaming, staff, guest Wi-Fi and services, two-cable EtherChannel uplinks, and Internet access through NAT.

You will use: 1 Multilayer Switch, 2 Access Switches, 16 Gaming PCs, 2 Wireless Access Points, 1 Edge Router, 1 ISP Router, 1 Public Server (plus 1 laptop or phone to test the Wi-Fi).

1. What you are building

Area	Devices	VLAN
Gaming Floor	16 wired Gaming PCs (8 per Access Switch)	10
Staff	Staff devices (no ports assigned in the notes yet)	20
Guest Wi-Fi	2 Wireless Access Points (one per Access Switch)	30
Services	Services VLAN (gateway only, no devices in the notes yet)	40
Network Cabinet	Multilayer Switch + 2 Access Switches + Edge Router	99 (management)
Internet side (simulated)	ISP Router + Public Server	outside the internal network

Note: The network is built in layers: Access Switches (end devices) - Multilayer Switch (VLAN gateways, DHCP, routing) - Edge Router (NAT) - ISP Router - Public Server.

Key devices

Device	Hostname	Role
Multilayer Switch	L3-SW1	VLAN gateways, DHCP, routing, spanning-tree root; connects the internal network to the Edge Router
Access Switch 1	Access-SW1	8 Gaming PCs + Access Point 1; 2-cable EtherChannel to L3-SW1
Access Switch 2	Access-SW2	8 Gaming PCs + Access Point 2; 2-cable EtherChannel to L3-SW1
Edge Router	Edge-R1	NAT (private to public addresses) and the link to the ISP
ISP Router	ISP-R1	Simulates the Internet provider
Public Server	Public Server	Simulates a server on the Internet

2. Network plan (cheat sheet)

VLAN	Name	Subnet	Gateway (L3-SW1)	Switch ports	Addressing
10	Gaming	192.168.10.0/24	192.168.10.1	Access-SW1/2 F0/3 - F0/10	DHCP (.21+)
20	Staff	192.168.20.0/24	192.168.20.1	none yet (reserved)	DHCP (.21+)
30	Guest-WIFI	192.168.30.0/24	192.168.30.1	Access-SW1/2 F0/11	DHCP (.21+)
40	Services	192.168.40.0/24	192.168.40.1	none yet (reserved)	no DHCP pool (static)
99	Network-Management	192.168.99.0/24	192.168.99.1	none (switch SVIs)	Static: Access-SW1 = .2; Access-SW2 = .3

First 20 addresses of each DHCP subnet (.1 - .20) are reserved for gateways, infrastructure devices and future static devices.

EtherChannels (LACP)

Port-channel	L3-SW1 ports	Access Switch ports	Mode
Po1	F0/1 - F0/2	Access-SW1 F0/1 - F0/2	LACP (active)
Po2	F0/3 - F0/4	Access-SW2 F0/1 - F0/2	LACP (active)

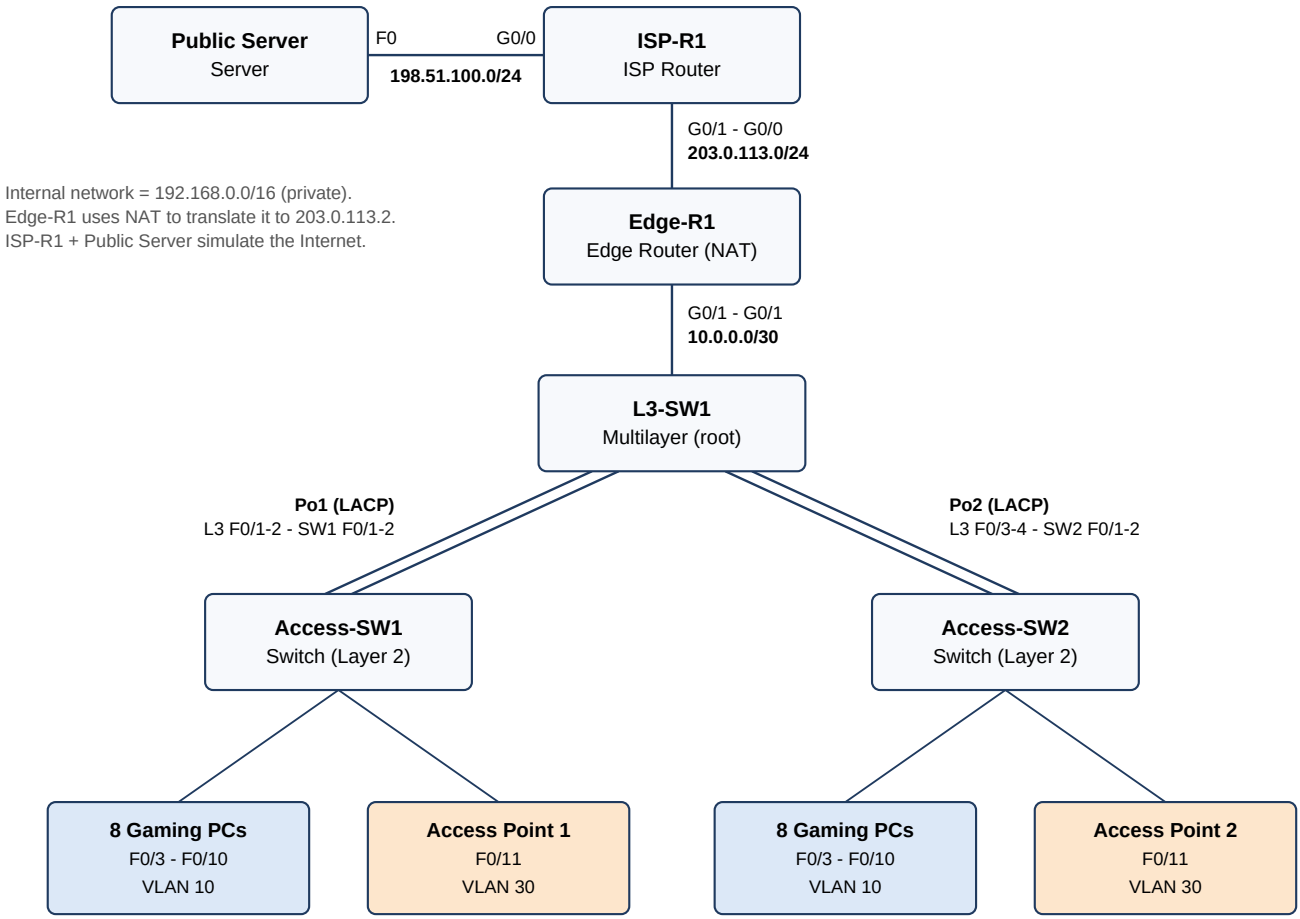
Note: An EtherChannel bundles 2 physical cables into 1 logical link: more bandwidth, a backup if one cable fails, and spanning tree does not block one of the two cables.

Routed links and routes

Where	Link / route	Details
L3-SW1 G0/1 to Edge-R1 G0/1	10.0.0.0/30	L3-SW1 = 10.0.0.2; Edge-R1 = 10.0.0.1 (NAT inside)
Edge-R1 G0/0 to ISP-R1 G0/1	203.0.113.0/24	Edge-R1 = 203.0.113.2 (NAT outside); ISP-R1 = 203.0.113.1
ISP-R1 G0/0 to Public Server	198.51.100.0/24	ISP-R1 = 198.51.100.1; Server = 198.51.100.10 (see section 7)
L3-SW1	Default route 0.0.0.0/0	Next hop 10.0.0.1 (Edge-R1)
Edge-R1	Route 192.168.0.0/16	Next hop 10.0.0.2 (L3-SW1)
Edge-R1	Default route 0.0.0.0/0	Next hop 203.0.113.1 (ISP-R1)

Note: 203.0.113.0/24 and 198.51.100.0/24 are example (documentation) address ranges that the lab uses as 'public' addresses.

Topology



3. Physical connections

#	From	To	Cable
1	ISP-R1 G0/0	Public Server F0	Ethernet
2	Edge-R1 G0/0	ISP-R1 G0/1	Ethernet
3	L3-SW1 G0/1	Edge-R1 G0/1	Ethernet
4	Access-SW1 F0/1	L3-SW1 F0/1	Ethernet
5	Access-SW1 F0/2	L3-SW1 F0/2	Ethernet
6	Access-SW2 F0/1	L3-SW1 F0/3	Ethernet
7	Access-SW2 F0/2	L3-SW1 F0/4	Ethernet
8	Gaming PCs 1 - 8	Access-SW1 F0/3 - F0/10	Ethernet
9	Gaming PCs 9 - 16	Access-SW2 F0/3 - F0/10	Ethernet
10	Access Point 1	Access-SW1 F0/11	Ethernet
11	Access Point 2	Access-SW2 F0/11	Ethernet

Note: Real life: the Access Points get power and network over one Ethernet cable using PoE (Power over Ethernet).

Tip: In Packet Tracer use copper straight-through cables between different devices and copper crossover between switches (or the automatic cable).

4. Multilayer switch configuration (L3-SW1)

Steps 1-2: Basic setup

Rename the switch and turn off DNS lookup.

```
enable
configure terminal
hostname L3-SW1
no ip domain-lookup
```

Step 3: Create the VLANs

Separate gaming, staff, guest Wi-Fi, services and management traffic.

```
vlan 10
 name Gaming
vlan 20
 name Staff
vlan 30
 name Guest-WIFI
vlan 40
 name Services
vlan 99
 name Network-Management
 exit
```

Step 4: EtherChannels (LACP)

Bundle F0/1 - F0/2 (to Access-SW1) into Port-channel 1 and F0/3 - F0/4 (to Access-SW2) into Port-channel 2. 'mode active' = LACP.

```
interface range f0/1 - 2
 channel-group 1 mode active
 exit
interface range f0/3 - 4
 channel-group 2 mode active
 exit
```

Step 5: Trunks on the Port-channels

Configure the logical Port-channels as trunks that allow only the VLANs we use ('interface po1' is the short form of 'interface port-channel 1').

```
interface port-channel 1
 description ## LACP TO Access-SW1 ##
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,99
 exit
interface port-channel 2
 description ## LACP TO Access-SW2 ##
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,99
 exit
```

> **Note:** If the switch model rejects the 'trunk encapsulation' line (it supports 802.1Q only), skip that line.

Step 6: Make L3-SW1 the Root Bridge

L3-SW1 becomes the spanning-tree Root Bridge for all VLANs.

```
spanning-tree vlan 10,20,30,40,99 root primary
```

Step 7: Routing and the SVIs (default gateways)

'ip routing' turns the switch into a router. Each SVI is the default gateway of its VLAN (SVIs are shut down by default, so 'no shutdown').

```
ip routing
interface vlan 10
 description ## Gaming Gateway ##
 ip address 192.168.10.1 255.255.255.0
 no shutdown
 exit
interface vlan 20
 description ## Staff Gateway ##
 ip address 192.168.20.1 255.255.255.0
 no shutdown
 exit
interface vlan 30
 description ## Guest-WIFI Gateway ##
 ip address 192.168.30.1 255.255.255.0
 no shutdown
 exit
interface vlan 40
 description ## Services Gateway ##
 ip address 192.168.40.1 255.255.255.0
 no shutdown
 exit
interface vlan 99
 description ## Network Management Gateway ##
 ip address 192.168.99.1 255.255.255.0
 no shutdown
 exit
```

Step 8: DHCP

L3-SW1 hands out IP addresses automatically. The first 20 addresses of each subnet are reserved for infrastructure devices. VLAN 40 and VLAN 99 have no pool.

```
ip dhcp excluded-address 192.168.10.1 192.168.10.20
ip dhcp excluded-address 192.168.20.1 192.168.20.20
ip dhcp excluded-address 192.168.30.1 192.168.30.20
! 1st pool - Gaming
ip dhcp pool GAMING
 network 192.168.10.0 255.255.255.0
 default-router 192.168.10.1
 dns-server 8.8.8.8
! 2nd pool - Staff
ip dhcp pool STAFF
 network 192.168.20.0 255.255.255.0
 default-router 192.168.20.1
 dns-server 8.8.8.8
! 3rd pool - Guest-WIFI
ip dhcp pool GUEST-WIFI
 network 192.168.30.0 255.255.255.0
 default-router 192.168.30.1
 dns-server 8.8.8.8
 exit
```

Step 9: SSH for secure remote management

SSH needs a hostname, a domain name, an RSA key and a local user. The domain name below is just an example.

```
ip domain-name gamingcenter.local
username admin privilege 15 secret YOUR_PASSWORD
crypto key generate rsa modulus 1024
ip ssh version 2
line vty 0 15
 login local
 transport input ssh
 exit
```

Step 10: Routed port to the Edge Router

G0/1 faces the Edge-R1. 'no switchport' turns it into a Layer 3 routed port with a /30 address.

```
interface g0/1
 description ## TO EDGE-R1 ##
 no switchport
 ip address 10.0.0.2 255.255.255.252
 no shutdown
 exit
```

Step 11: Default route

Traffic for any unknown destination goes to the Edge Router.

```
ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

Step 12: Guest ACL (Internet only)

Protect the internal network from the Guest Wi-Fi. The switch routes between all VLANs, so block guests from every internal 192.168.x.x network.

```
ip access-list extended GUEST-FILTER
 permit udp any eq bootpc any eq bootps
 deny ip 192.168.30.0 0.0.0.255 192.168.0.0 0.0.255.255
 permit ip 192.168.30.0 0.0.0.255 any
 exit
interface vlan 30
 ip access-group GUEST-FILTER in
 exit
```

ACL line	What it does
permit udp ... bootpc ... bootps	Lets guests get an IP by DHCP (UDP 68 = DHCP client, UDP 67 = DHCP server)
deny ... 192.168.0.0 0.0.255.255	Blocks guests from every internal network
permit ip 192.168.30.0 ... any	Allows everything else = Internet only

> **Note:** The DHCP rule must come first, otherwise guests cannot get an IP address.

Step 13: Save the configuration

So it survives a power loss ('write memory' does the same).

```
end
copy running-config startup-config
```

5a. Access switch 1 configuration (Access-SW1)

Steps 1-2: Basic setup and VLANs

Rename the switch and create the same VLANs that were created on L3-SW1.

```
enable
configure terminal
hostname Access-SW1
vlan 10
 name Gaming
vlan 20
 name Staff
vlan 30
 name Guest-WIFI
vlan 40
 name Services
vlan 99
 name Network-Management
 exit
```

Step 3: Gaming PC ports (VLAN 10)

F0/3 - F0/10 connect the 8 Gaming PCs. PortFast makes end-device ports come up fast.

```
interface range f0/3 - 10
 description ## Gaming PCs ##
 switchport mode access
 switchport access vlan 10
 spanning-tree portfast
 exit
```

Step 4: Access Point port (VLAN 30)

```
interface f0/11
 description ## Guest-WIFI ##
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 exit
```

Step 5: EtherChannel (LACP)

F0/1 - F0/2 go to L3-SW1 and form Port-channel 1.

```
interface range f0/1 - 2
 channel-group 1 mode active
 exit
```

Step 6: Trunk on the Port-channel

Make the Port-channel a trunk, then check that the bundle is up.

```
interface port-channel 1
 description ## LACP TO L3-SW1 ##
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30,40,99
 exit
do show etherchannel summary
```

Step 7: Management SVI and default gateway

The switch gets its own static address in VLAN 99.

```
interface vlan 99
 ip address 192.168.99.2 255.255.255.0
 no shutdown
 exit
ip default-gateway 192.168.99.1
```

Step 8: SSH for secure remote management

Same commands as on L3-SW1.

```
ip domain-name gamingcenter.local
username admin privilege 15 secret YOUR_PASSWORD
crypto key generate rsa modulus 1024
ip ssh version 2
line vty 0 15
 login local
 transport input ssh
 exit
```

Step 9: Save the configuration

```
end
copy running-config startup-config
```

5b. Access switch 2 configuration (Access-SW2)

Repeat steps 1-9 of Access-SW1. Only these items change:

Item	Access-SW1	Access-SW2
Hostname	hostname Access-SW1	hostname Access-SW2
EtherChannel (step 5)	channel-group 1 mode active	channel-group 2 mode active
Port-channel (step 6)	interface port-channel 1	interface port-channel 2
Management IP (step 7)	192.168.99.2 255.255.255.0	192.168.99.3 255.255.255.0

Note: Everything else is identical: the VLANs, ports F0/3 - F0/10 in VLAN 10, F0/11 in VLAN 30, the trunk settings, the default gateway 192.168.99.1 and SSH.

6. Edge router configuration (Edge-R1)

Steps 1-2: Basic setup

Rename the router and turn off DNS lookup.

```
enable
configure terminal
hostname Edge-R1
no ip domain-lookup
```

Step 3: Interface to the L3 Switch (NAT inside)

G0/1 faces L3-SW1: 10.0.0.1 /30, and it is the inside of the NAT boundary.

```
interface g0/1
 description ## TO L3-SW1 ##
 ip address 10.0.0.1 255.255.255.252
 ip nat inside
 no shutdown
 exit
```

Step 4: Interface to the ISP (NAT outside)

G0/0 faces ISP-R1: 203.0.113.2 /24, and it is the outside of the NAT boundary.

```
interface g0/0
 description ## To ISP-R1 ##
 ip address 203.0.113.2 255.255.255.0
 ip nat outside
 no shutdown
 exit
```

Step 5: Routes

One route to the internal networks (through L3-SW1) and one default route to the ISP router.

```
! internal networks route
ip route 192.168.0.0 255.255.0.0 10.0.0.2
! ISP router route (default route)
ip route 0.0.0.0 0.0.0.0 203.0.113.1
```

Step 6: NAT (private to public addresses)

Private 192.168.x.x addresses cannot be used on the Internet. The ACL selects the internal networks and NAT translates them to the router's outside address.

```
access-list 1 permit 192.168.0.0 0.0.255.255
ip nat inside source list 1 interface g0/0 overload
```

Command	What it does
ip nat inside / ip nat outside	Marks the internal side and the Internet side of the router
access-list 1 permit 192.168.0.0 ...	Selects which addresses are translated (all internal networks)
ip nat inside source list 1 interface g0/0 overload	Translates them to the address of G0/0; many PCs share one public address

Step 7: Save the configuration

```
end
copy running-config startup-config
```

7. ISP router and Public Server configuration

ISP-R1: rename and enable both interfaces

G0/0 faces the Public Server and G0/1 faces Edge-R1.

```
enable
configure terminal
hostname ISP-R1
interface g0/0
 description ## To Public Server ##
 ip address 198.51.100.1 255.255.255.0
 no shutdown
 exit
interface g0/1
 description ## To Edge-R1 ##
 ip address 203.0.113.1 255.255.255.0
 no shutdown
 exit
end
copy running-config startup-config
```

Note: ISP-R1 needs no static routes: both of its networks are directly connected.

Public Server: static IP settings

Setting	Value
IPv4 address	198.51.100.10 (static, assumed - see warning)
Subnet mask	255.255.255.0 (/24)
Default gateway	198.51.100.1

Warning: The notes give the server 203.0.113.10 /24 with gateway 203.0.113.1, but the server is cabled to ISP-R1 G0/0 (198.51.100.1) and 203.0.113.1 is already on ISP-R1 G0/1 (the link to Edge-R1). The same /24 cannot be on two interfaces of one router, so this guide puts the server in 198.51.100.0/24. The host address .10 is an assumption - use what your video shows if it is different.

8. End devices

Device	VLAN	IP address	Gateway	DNS
Gaming PCs (16)	10	DHCP (automatic)	192.168.10.1	8.8.8.8
Guest devices (via the APs)	30	DHCP (automatic)	192.168.30.1	8.8.8.8
Public Server	-	198.51.100.10 (static)	198.51.100.1	-

Access Points (both): open the configuration and set an SSID and a key-phrase (Wi-Fi password).

Test 1: connect a device to each Access Point and check that it gets an IP address by DHCP (192.168.30.x).

Test 2: check that all 16 Gaming PCs got an IP address by DHCP (192.168.10.x).

> **Note:** VLAN 20 (Staff) and VLAN 40 (Services) have no devices yet; their gateways exist on L3-SW1 and Staff already has a DHCP pool.

9. Quick verification

Check	Command / test
VLANs and port assignment	show vlan brief (L3-SW1, Access-SW1, Access-SW2)
EtherChannels up	show etherchannel summary (Po1 and Po2 show (SU), member ports show (P))
Trunks up, VLANs allowed	show interfaces trunk
L3-SW1 is the Root Bridge	show spanning-tree vlan 10 (on L3-SW1: 'This bridge is the root')
SVIs and routed port up	show ip interface brief (L3-SW1)
Routing	show ip route (L3-SW1: default via 10.0.0.1; Edge-R1: 192.168.0.0/16 via 10.0.0.2, default via 203.0.113.1)
DHCP pools and leases	show ip dhcp pool | show ip dhcp binding (L3-SW1)
Guest ACL applied	show access-lists | show ip interface vlan 30
NAT working	show ip nat translations (Edge-R1)
SSH ready	show ip ssh (L3-SW1, Access-SW1, Access-SW2)
Gaming PC works	ipconfig shows 192.168.10.x (.21 or higher); ping 192.168.10.1 (gateway)
Internet works	From a Gaming PC: ping 198.51.100.10 (the Public Server)
Guest isolation works	From a guest device: ping 192.168.10.1 should FAIL; the Public Server should work

Appendix: corrections made to the original notes

In the notes	Used in this guide
'Access-SW3' interfaces F0/3 - F0/10 to 8 PCs	Access-SW2
'interface vlan 120' (Staff gateway)	interface vlan 20
'192.168.2O.0' and '192.168.3O.0' (letter O)	192.168.20.0 and 192.168.30.0 (zeros)
'ip dhcp excluded-addresses'	ip dhcp excluded-address
L3-SW1 G0/1 '10.0.0.2 255.255.255.0' (notes say /30)	10.0.0.2 255.255.255.252
'ip route 192.168.0.0 255.255.0.0. 10.0.0.2' (extra dot)	ip route 192.168.0.0 255.255.0.0 10.0.0.2
VLANs created with 'description ## Gaming ##'	name Gaming (VLANs use 'name')
'##' comments after commands (for example '## ISP router route ##')	'!' comment lines
'crypto key generate rsa' then '1024' on a new line	crypto key generate rsa modulus 1024
SSH steps on the Access Switches titled 'Multilayer Switch'	Titled Access-SW1 / Access-SW2
Password 'Cisco123'	YOUR_PASSWORD (use your own password)
Public Server '203.0.113.10 /24', gateway '203.0.113.1'	198.51.100.10 /24, gateway 198.51.100.1 (assumption - see section 7)

Added so the commands work (not written in the notes): 'switchport mode access' and 'switchport access vlan' on the access ports, 'ip ssh version 2', 'exit' lines, 'copy running-config startup-config' (the notes use 'write memory'), the Access-SW2 differences table, the verification commands, and the address of the Public Server.